TRƯỜNG ĐẠI HỌC
CÔNG NGHỆ KỸ THUẬT TP. HỒ CHÍ MINH
Ho Chi Minh City University of Technology and Engineering
HCM-UTE

LECTURE 2

Basic CMOS Device Physics

Design of Analog CMOS Integrated Circuits

Department of Electronics and Information Engineering | Ho Chi Minh City University of Technology and Engineering (HCM-UTE), Vietnam

Based on Design of Analog CMOS Integrated Circuits, Chapter 2 (Razavi)

BẢNG TÓM TẮT — TIẾNG VIỆT

So sánh Analog và Digital IC Design theo cách dùng CMOS transistor

Tiêu chí	Digital IC Design	Analog IC Design
Vai trò của transistor	Công tắc nhị phân (ON / OFF)	Bộ khuếch đại điện áp-thành-dòng liên tục
Vùng hoạt động chủ đạo	Cutoff và triode sâu	Saturation (hoặc subthreshold khi công suất thấp)
Vai trò vùng saturation	Chỉ đi qua thoáng qua, không quan trọng	Vùng làm việc chính, cần chính xác
Đại lượng thiết kế quan tâm	R_on, tổng điện dung, margin của VTH	gm, ro, gmb, CGS (fT), nhiễu, độ tuyến tính
Khái niệm tín hiệu	Full-swing, rail-to-rail (large-signal)	Dao động nhỏ quanh điểm Q (small-signal)
Điểm làm việc DC (bias)	Không phải khái niệm có ý nghĩa	Bắt buộc — cần mạch phân cực / gương dòng
Độ nhạy với mismatch	Chủ yếu ảnh hưởng tốc độ/công suất, không sai logic	Trực tiếp gây offset voltage và sai số gain
Ảnh hưởng điều biến chiều dài kênh (λ)	Không đáng kể	Quyết định trực tiếp ro và gain nội tại gm·ro
Xu hướng theo scaling công nghệ	Thường được lợi: nhanh hơn, tiết kiệm điện hơn	Thường bị thiệt: giảm headroom, giảm gm·ro
Mục tiêu thiết kế	Đúng chức năng + tốc độ/công suất/diện tích	Độ trung thực tín hiệu: gain, tuyến tính, nhiễu, băng thông

SUMMARY TABLE — ENGLISH

Analog vs. Digital IC Design: Using the CMOS Transistor

Criterion	Digital IC Design	Analog IC Design
Role of the transistor	Binary switch (ON / OFF)	Continuous voltage-to-current amplifier
Dominant operating region	Cutoff and deep triode	Saturation (or subthreshold for low power)
Role of the saturation region	Brief transition only, not critical	Primary operating region, must be precise
Key design parameters	RON, total capacitance, VTH margin	gm, ro, gmb, CGS (fT), noise, linearity
Signal concept	Full-swing, rail-to-rail (large-signal)	Small perturbation around a Q-point (small-signal)
DC operating point (bias)	Not a meaningful concept	Essential — needs bias circuit / current mirror
Sensitivity to mismatch	Mainly affects speed/power, not logic correctness	Directly causes offset voltage and gain error
Effect of channel-length modulation (λ)	Negligible	Directly sets ro and the intrinsic gain gm·ro
Trend with technology scaling	Generally benefits: faster, lower power	Often penalized: less headroom, lower gm·ro
Design goal	Functional correctness + speed/power/area	Signal fidelity: gain, linearity, noise, bandwidth



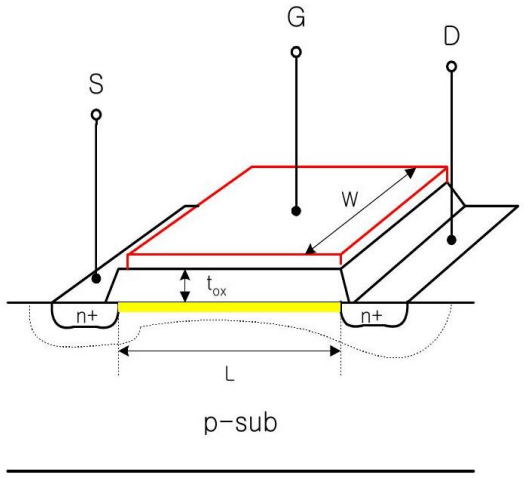
MOS Current-Voltage Model

- MOS structure, terminals, and charge-drift current model
- Triode and saturation region derivations
- Transconductance (gm) and the NMOS/PMOS boundary condition

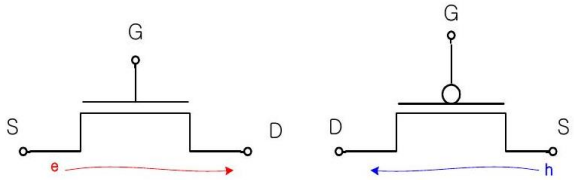
MOS Structure and Terminal Definitions

- Gate (polysilicon) sits above a thin oxide layer between two n+ regions — drain (D) and source (S).
- Channel width W and length L are the key geometric design parameters.
- NMOS conducts via electrons (no bubble at gate); PMOS conducts via holes (bubble at gate).
- For NMOS, the higher-voltage terminal is the drain; the lower one is the source.

❑ MOS structure



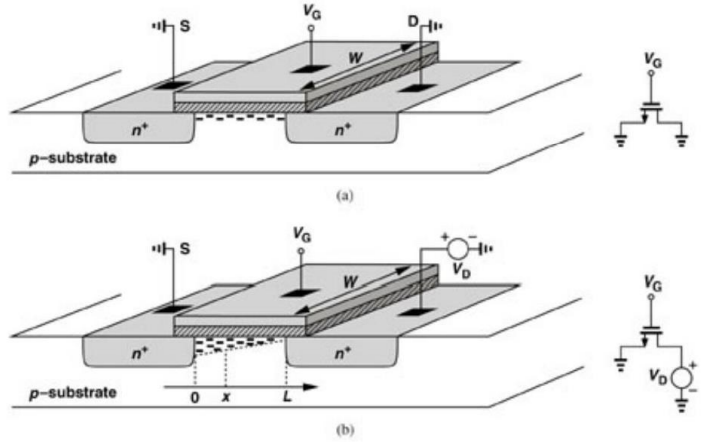
❑ MOS terminals



■ For NMOS, Higher biased node is the drain and lower biased node is the source

Charge-Drift Current Model

- At VDS = 0, gate voltage VG forms a uniform inversion (channel) charge along the whole channel.
- Adding VD > 0 makes the channel voltage V(x) rise from 0 (at S) to VDS (at D).
- Channel charge density Qd(x) therefore decreases from source to drain — the channel is "tapered."



$$I = Q_d \cdot v$$

$$Q_d = WC_{ox}(V_{GS} - V_{TH})$$

$$Q_d(x) = WC_{ox}(V_{GS} - V(x) - V_{TH})$$

Triode-Region Current: Derivation

- Drift current I = Qd·v, with v = μE and E(x) = −dV/dx.
- Integrating ID over the channel length (0→L) and V (0→VDS) removes the position dependence.
- Result: the triode-region current equation, ID = μnCox(W/L)[(VGS−VTH)VDS − ½VDS²].

$$I_D = |WC_{ox}[V_{GS} - V(x) - V_{TH}]v|$$

$$Given \quad v = \mu E \quad and \quad E(x) = -\frac{dV(x)}{dx}$$

$$I_D = WC_{ox}[V_{GS} - V(x) - V_{TH}]\mu_n \frac{dV(x)}{dx}$$

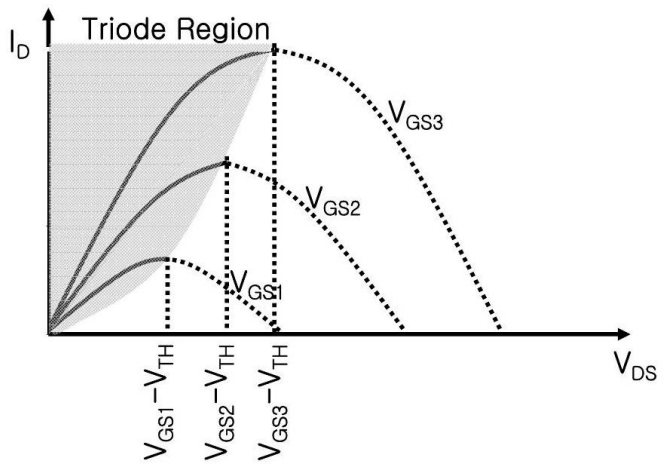
$$\int_{x=0}^{L} I_D dx = \int_{V=0}^{VDS} WC_{ox}\mu_n[V_{GS} - V(x) - V_{TH}]dV$$

$$I_D = \mu_n C_{ox}\frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

Triode-Region I-V Characteristics

- For each fixed VGS, ID rises almost linearly at small VDS, then bends as the VDS² term grows.
- Each curve's peak occurs exactly at VDS = VGS − VTH — the onset of pinch-off.
- Dashed extensions show the (invalid) triode formula beyond this boundary.

$$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$



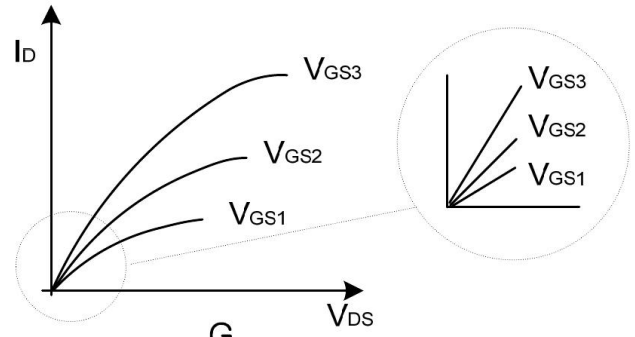
Deep Triode Region and ON-Resistance

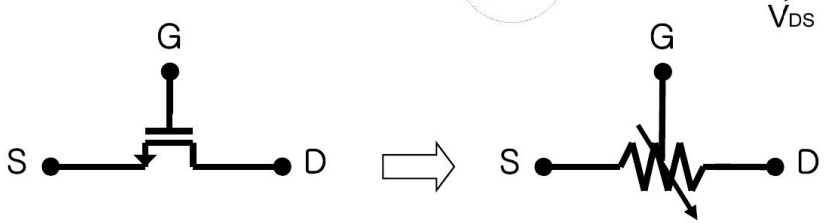
- When **VDS ≪ 2(VGS−VTH)**, the VDS² term is negligible: ID ≈ μnCox(W/L)(VGS−VTH)VDS.
- The transistor behaves like **a linear resistor**: RON = 1/[μnCox(W/L)(VGS−VTH)].
- Basis for using a MOSFET as an analog switch (e.g., sample-and-hold, switched-capacitor circuits).

$$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

$$I_D = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})V_{DS}, \quad V_{DS} << 2(V_{GS} - V_{TH})$$

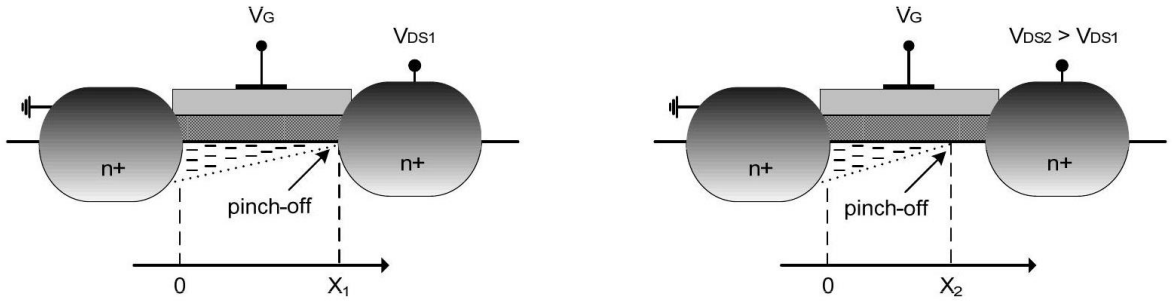
$$R_{ON} = \frac{1}{\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})}$$





Pinch-Off and the Saturation Region

- Beyond pinch-off, the channel voltage at the pinch-off point is fixed at VGS − VTH, independent of VDS.
- Substituting VDS = VGS−VTH into the triode equation gives the square-law saturation current.
- ID = (μnCox/2)(W/L)(VGS−VTH)² — nearly independent of VDS, so the MOSFET acts as a current source.



$$V(x_1) = V_{GS} - V_{TH} \qquad V(x_2) = V_{GS} - V_{TH}$$

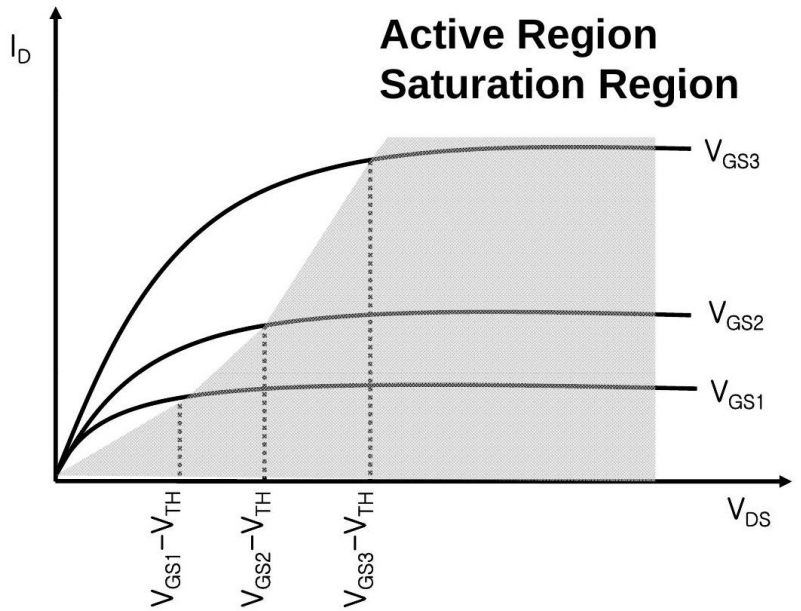
$$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

$$V'_{DS} = V_{GS} - V_{TH} \quad (Pinch-off)$$

$$I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{TH})^2$$

Triode and Saturation I-V Characteristics

- Each VGS curve transitions from the triode (rising) region into the flat saturation region.
- The shaded parabola is the pinch-off locus (VDS = VGS−VTH), separating the two regions.
- This combined I-V map is the standard reference characteristic for MOSFET operation.

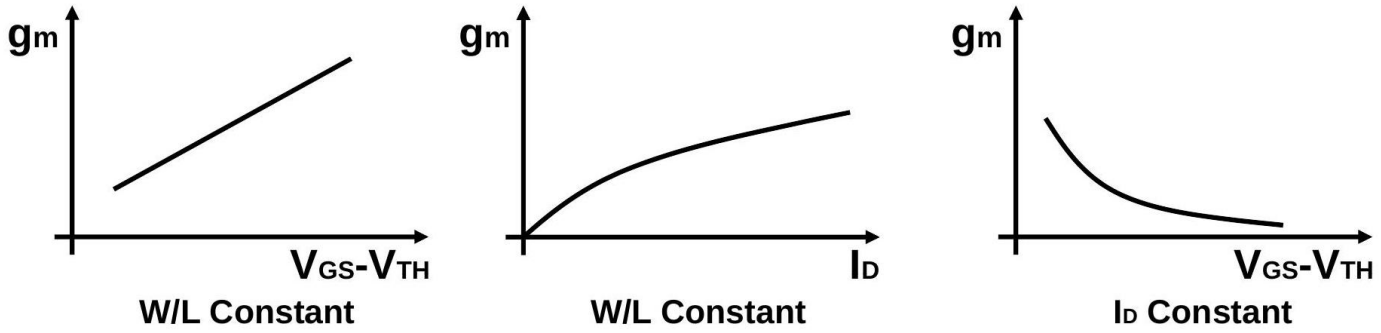


Transconductance (gm)

- gm = ∂ID/∂VGS at fixed VDS — measures **how sensitively ID responds to VGS**, setting amplifier gain.
- Two equivalent forms: gm = √(2μnCox(W/L)ID) and gm = 2ID/(VGS−VTH).
- For fixed ID, a smaller overdrive (VGS−VTH) gives higher gm — basis of low-power gm/ID design.

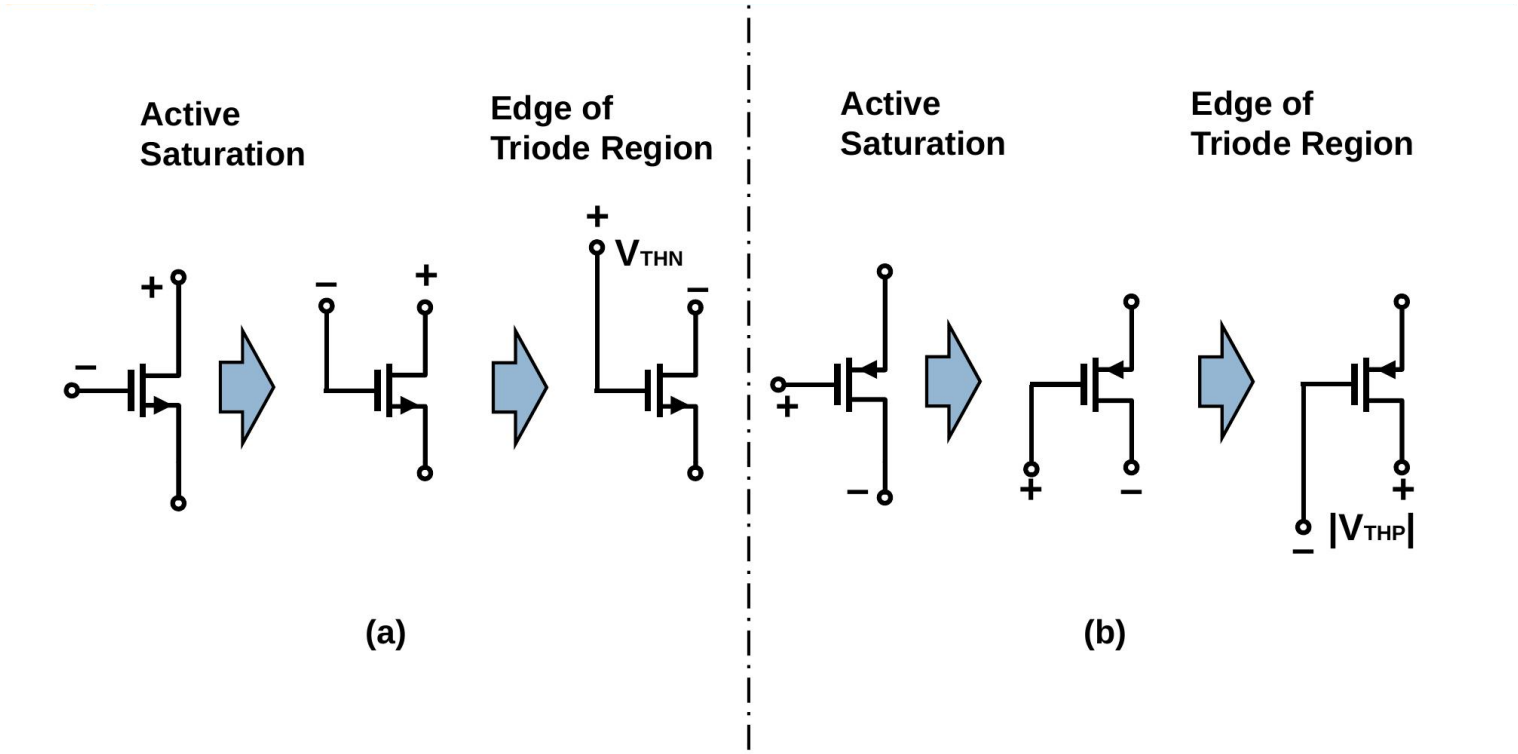
$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{VDS\ \ constant} = \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})$$

$$g_m = \sqrt{2\mu_n C_{ox}\frac{W}{L} I_D} = \frac{2I_D}{V_{GS}-V_{TH}}$$



Saturation Boundary: NMOS vs. PMOS

- The triode/saturation boundary occurs at VDS = VGS − VTH for NMOS.
- For PMOS, all voltage/current directions reverse; the boundary uses |VTHP|.
- Essential for analyzing CMOS circuits (logic gates, push-pull stages, current mirrors).



II

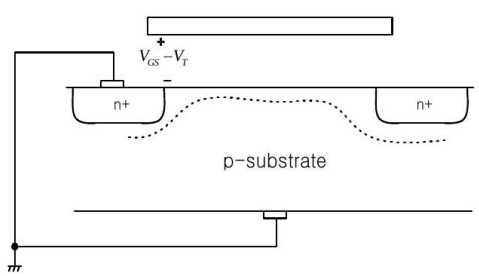
Threshold Voltage

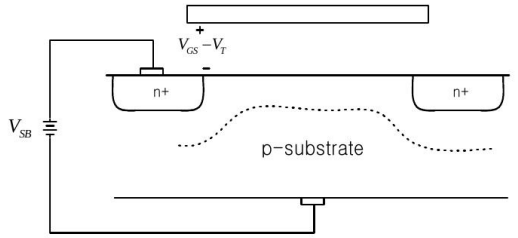
- Body effect: concept and physical origin
- Threshold voltage components (ΦMS, 2ΦF, Qdep/Cox)
- Body-effect impact on circuit behavior

Body Effect: Concept and Basic Equation

- With VSB = 0, the body/substrate is tied to the source — no body effect.
- Adding VSB reverse-biases the body-source junction, changing the threshold voltage.
- VT = VT0 + γ(√|2ΦF+VSB| − √|2ΦF|) — **VT increases as VSB increases**
  (VT0 = VT(VSB=0))

❑ Body effect





$$V_T = V_{TO} + \gamma\left(\sqrt{|2\Phi_F + V_{SB}|} - \sqrt{|2\Phi_F|}\right)$$

Threshold Voltage: Physical Components

- **VTH(VT0)** = ΦMS + 2ΦF + Qdep/Cox — three physical contributions (VSB=0 i.e., no body effect)
- ΦMS: work-function difference between the gate material and the silicon substrate.
- 2ΦF: voltage needed to bring the surface into strong inversion (forms the channel).
- Qdep/Cox: voltage dropped across the oxide due to depletion charge under the gate.

$$V_{TH} = \Phi_{MS} + 2\Phi_F + \frac{Q_{dep}}{C_{ox}}, \quad where$$

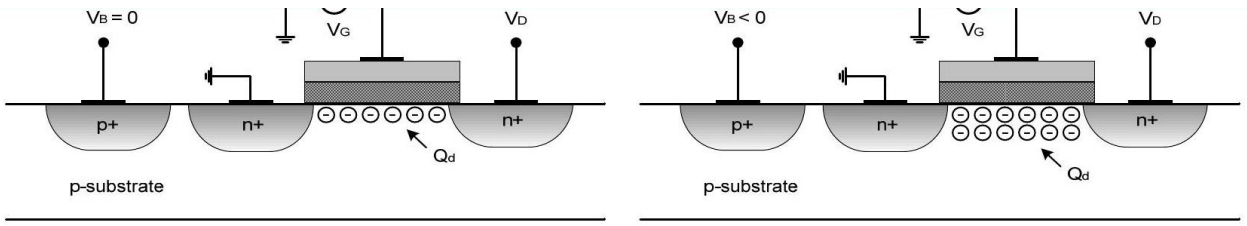
$$\Phi_{MS} = \Phi_{gate} - \Phi_{silicon}$$

$$\Phi_F = \left(kT/q\right)\ln\left(N_{sub}/n_i\right)$$

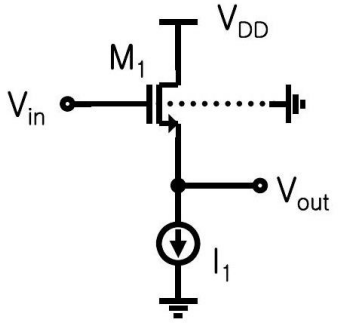
$$Q_{dep} = \sqrt{4q\varepsilon_{si}|\Phi_F|N_{sub}}$$

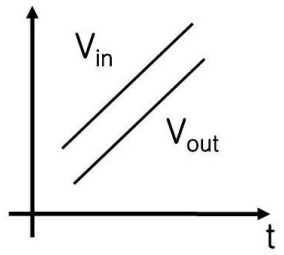
Body Effect Mechanism and Circuit Impact

- A more negative body voltage VB widens the depletion region, increasing VTH.
- $\gamma = \sqrt{(2q\varepsilon_{si}N_{sub})}/C_{ox}$ — thinner oxide or lighter doping reduces the body effect.
- In a source follower, VSB varies with Vout, so VTH shifts — bending the Vin–Vout transfer curve.

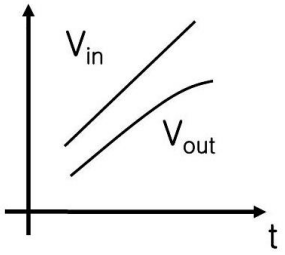


$$V_{TH} = V_{TH0} + \gamma\left(\sqrt{2\Phi_F + V_{SB}} - \sqrt{2\Phi_F}\right), \quad \gamma = \frac{\sqrt{2q\varepsilon_{si}N_{sub}}}{C_{ox}}$$





No Body Effect



With Body Effect

III

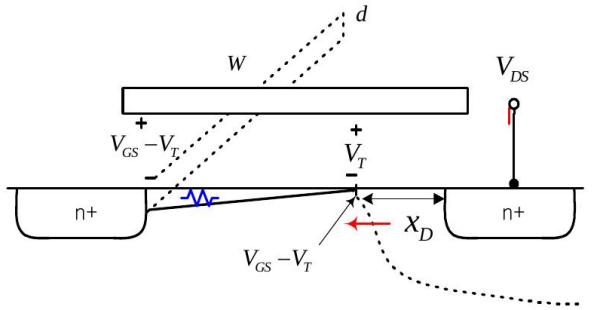
Channel-Length Modulation

- Physical origin of the (1+λVDS) correction
- Mathematical derivation of λ
- Updated I-V characteristics and gm

Channel-Length Modulation: Physical Origin

- Beyond pinch-off, the pinch-off point recedes toward the source as VDS increases, shortening the channel.
- This adds a correction factor: ID = (μCox/2)(W/L)(VGS−VTH)²(1+λ VDS).
- λ ≈ (1/L)(1/√Na): **channel-length modulation coefficient**, longer channels and heavier substrate doping reduce the effect.

❑ Channel−length modulation effect

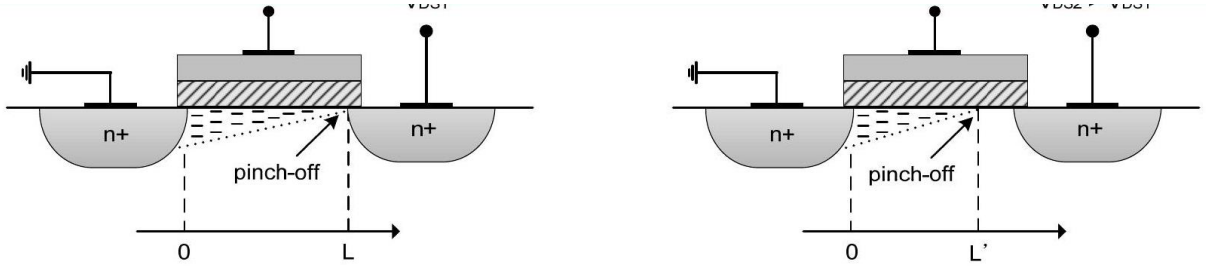


$$I = \frac{\mu C_{ox}}{2}\left(\frac{W}{L}\right)(V_{GS}-V_T)^2(1+\lambda V_{DS})$$

$$\lambda \approx \frac{1}{L}\frac{1}{\sqrt{N_a}}$$

Deriving the (1+λVDS) Correction Factor

- **Define the shortened effective length** $L' = L - \Delta L$.
- First-order approximation: $1/L' \approx (1/L)(1 + \Delta L/L) = (1/L)(1 + \lambda VDS)$.
- Substituting L' for L in the square-law equation yields the (1+λVDS) correction term.



$V(x_1) = V_{GS} - V_{TH}$ $\qquad$ $V(x_2) = V_{GS} - V_{TH}$

$$L' = L - \Delta L$$

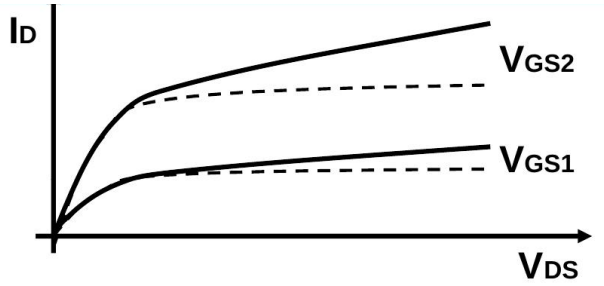
$$1/L' = \frac{1}{L}(1 + \Delta L/L)$$

$$1/L' = \frac{1}{L}(1 + \lambda V_{DS}), \quad \lambda V_{DS} = \Delta L/L$$

$$I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{TH})^2(1 + \lambda V_{DS})$$

I-V Characteristics with Channel-Length Modulation

- **With λ included**, ID in saturation rises slightly with VDS instead of staying flat.
- Extrapolated backward, curves cross the axis at VDS = −1/λ — analogous to the BJT's Early voltage.
- Updated transconductance: gm = μnCox(W/L)(VGS−VTH)(1+λVDS).



$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH})(1 + \lambda VDS)$$

$$g_m = \sqrt{2\mu_n C_{ox} W/L \, I_D}$$



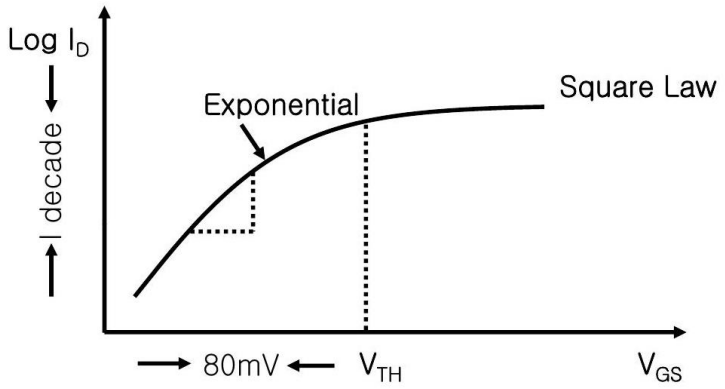
Subthreshold Current

- Exponential I-V behavior below VTH

Subthreshold Conduction: Exponential I-V Behavior

- Even for VGS < VTH, a small but **nonzero current still flows**.
- ID = I0·exp[VGS/(ζkT/q)] — an exponential function of VGS, unlike the square law above VTH.
- Relevant to ultra-low-power digital leakage and to subthreshold (weak-inversion) analog design.

$$I_D = I_0 \exp\left(\frac{V_{GS}}{\zeta kT/q}\right)$$



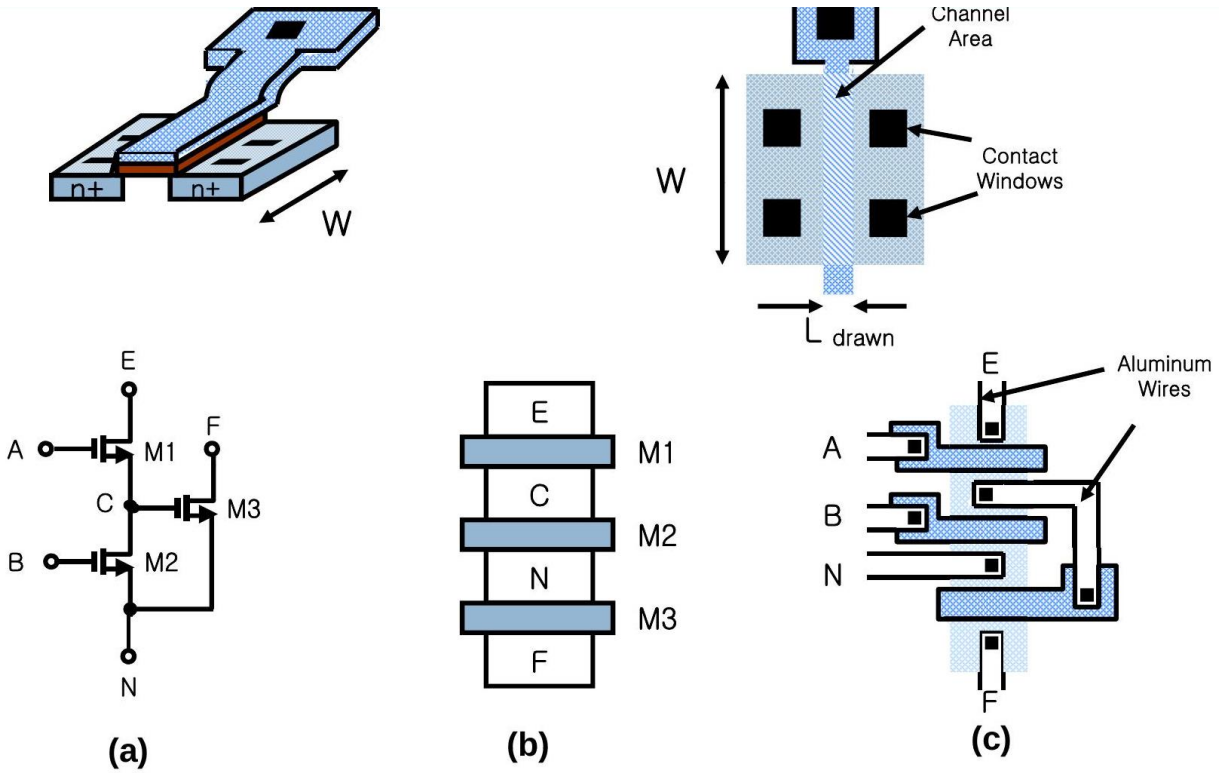
V

Capacitance-Voltage Model

- MOSFET layout and parasitic capacitance components
- Capacitance vs. VGS characteristics (C1–C6)
- CGS derivation and the MOS C–V curve

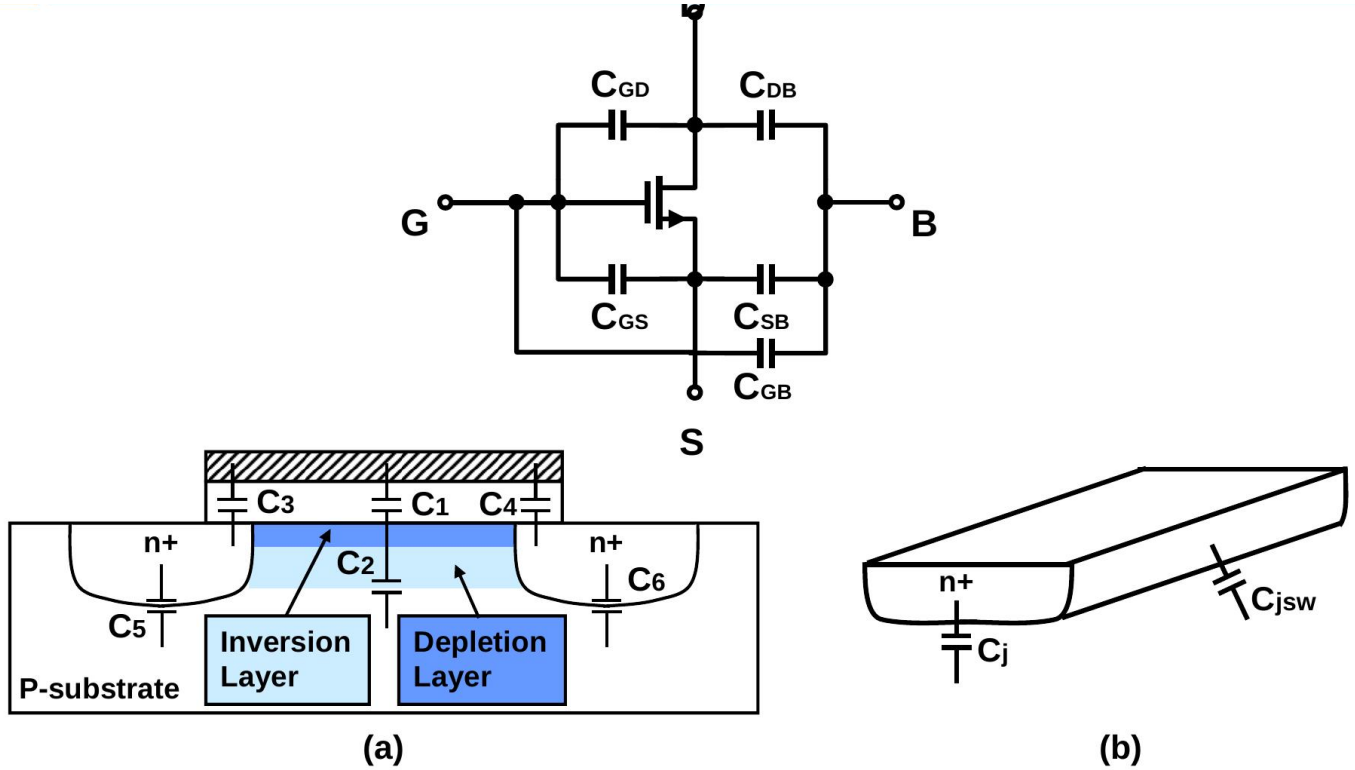
MOSFET Layout: From Schematic to Silicon

- Layout translates the abstract circuit schematic into physical geometry on silicon.
- Key features: channel area, contact windows, drawn length (Ldrawn) vs. effective channel length.
- Series-connected transistors can share diffusion regions to save area — shown at 3 abstraction levels.



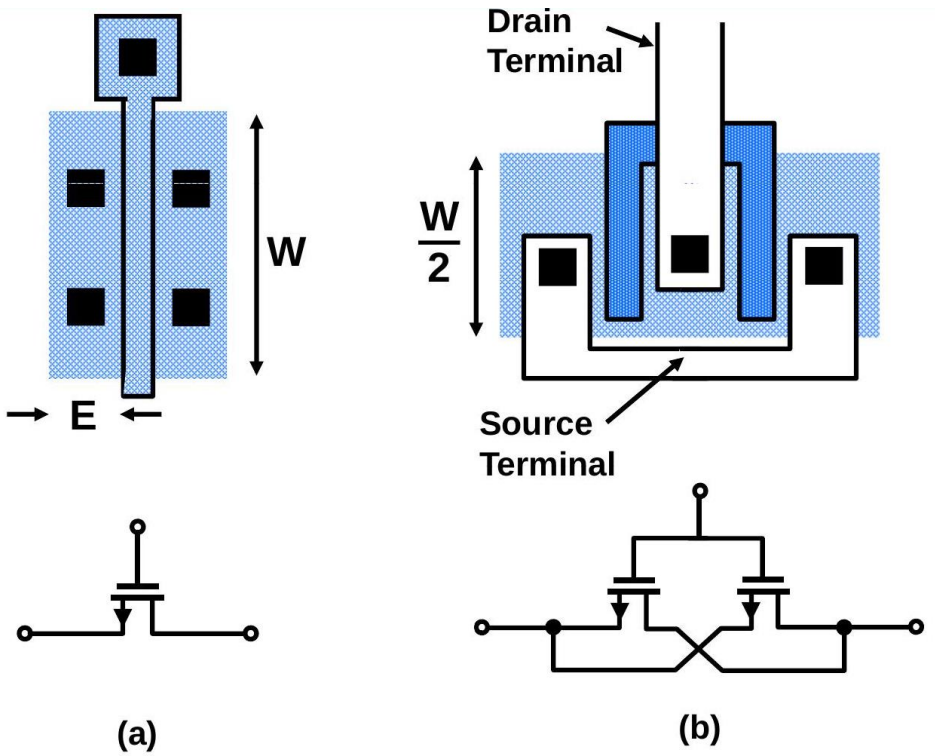
Parasitic Capacitance Components

- Five capacitances connect the four terminals: CGD, CGS, CGB, CDB, CSB.
- Gate-related capacitance (C1–C4) comes from the gate-channel/overlap regions.
- Junction capacitance (C5, C6) comes from the source/drain-to-body p-n junctions: Cj (bottom) + Cjsw (sidewall).



Layout Technique: Folded Structure to Reduce CDB

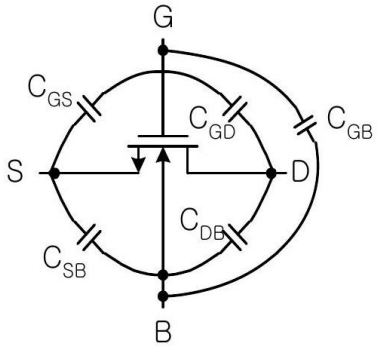
- A single-finger layout (a) has one large drain diffusion region — larger CDB.
- Splitting into two half-width fingers (b) sharing a common source shrinks the drain area and perimeter.
- Smaller drain diffusion means lower Cj and Cjsw — important for high-speed/RF layout.



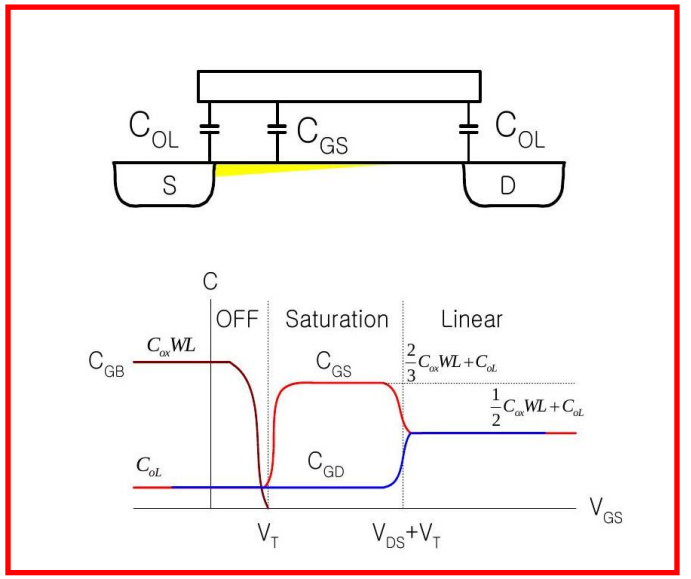
MOS Capacitance vs. VGS: Off, Saturation, Triode

- In cutoff, CGB is maximum (≈CoxWL); CGS and CGD are only the small overlap capacitance.
- In saturation, pinch-off shields the gate from the body: CGB drops to ~0; CGS rises to (2/3)CoxWL + COL.
- In the triode region, the channel is continuous and CGS = CGD = (1/2)CoxWL + COL.

❑ MOS capacitances



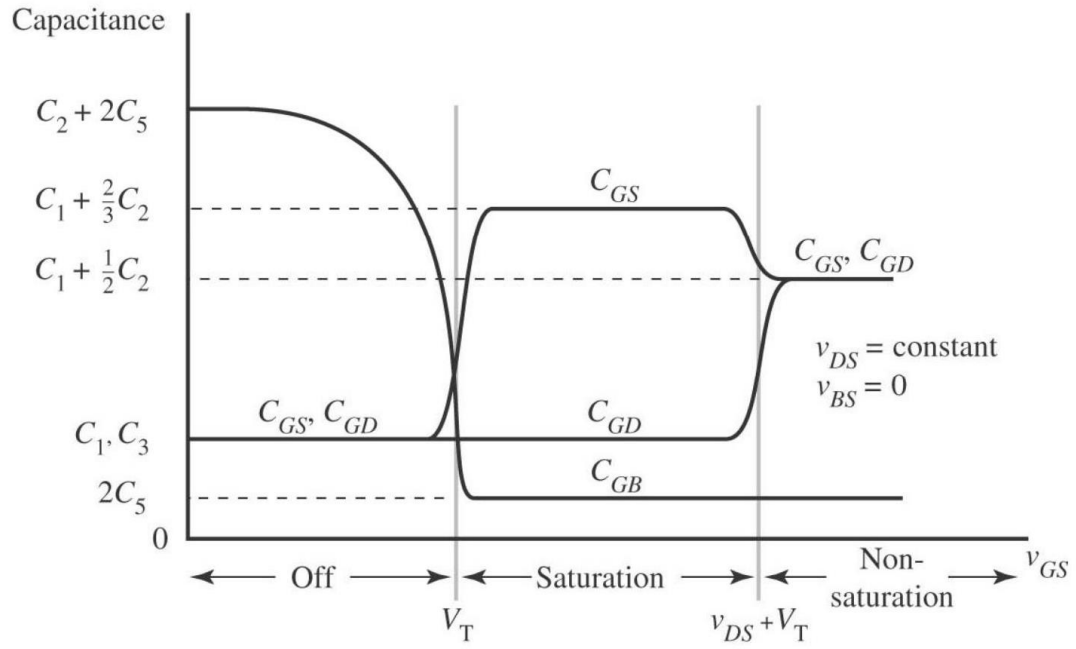
$$C_j = \frac{C_{jo}}{\left(1+\frac{V_{BIAS}}{V_{BI}}\right)^{\frac{1}{2}}}$$



- C_{OL}: overlap capacitance
- C_{SB}, C_{DB}: junction capacitance

Capacitance–VGS Characteristics (C1–C6)

- Same three regions (Off, Saturation, Non-saturation), now labeled with the C1–C6 layout components.
- Boundaries occur at VGS = VTH (channel forms) and VGS = VDS + VTH (saturation/triode boundary).
- Links each capacitance's physical origin (previous slide) to its observed value here.



Deriving CGS in Saturation (Part 1)

- Starts from the same current equation used for the triode-region derivation.
- Total channel charge QT = WCox∫[VGS−V(x)−VTH]dx, converted to an integral over V via dx/dV.
- Integrating from V=0 to V=VGS−VTH (the pinch-off voltage) sets up the final expression for QT.

$$I_D = WC_{ox}[V_{GS} - V(x) - V_{TH}]\mu_n \frac{dV(x)}{dx}$$

$$Q_T = WC_{ox}\int_0^L [V_{GS} - V(x) - V_{TH}]dx$$

$$= WC_{ox}\int_0^L [V_{GS} - V(x) - V_{TH}]dV \frac{dx}{dV}$$

$$Q_T = \frac{W^2C_{ox}^2\mu}{I_d}\int_0^{VGS-VTH} [V_{GS} - V(x) - V_{TH}]^2 dV = \frac{W^2C_{ox}^2\mu}{I_d}\frac{(V_{GS} - V_{TH})^3}{3}$$

$$I_D = \frac{W}{2L}C_{ox}[V_{GS} - V_{TH}]^2\mu_n$$

Deriving CGS in Saturation (Part 2): CGS = (2/3)CoxWL

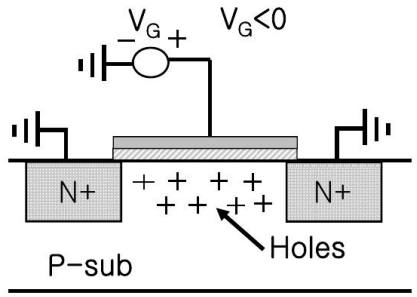
- Substituting ID = (W/2L)Cox(VGS−VTH)²μn simplifies QT to (2/3)WLCox(VGS−VTH).
- The 2/3 factor reflects the channel's tapered shape — thick at the source, pinched to zero at the drain.
- CGS = ∂QT/∂VGS = (2/3)WLCox — dominates the transistor's cutoff frequency fT.

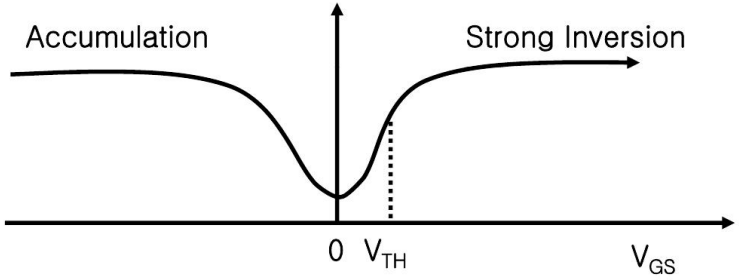
$$Q_T = \frac{2}{3} WLC_{OX} (V_{GS} - V_{TH})$$

$$C_{GS} = \frac{\partial Q_T}{\partial V_{GS}} = \frac{2}{3} WLC_{OX}$$

MOS Capacitor C–V Curve

- At very negative VGS, holes accumulate at the surface — capacitance is near Cox.
- Near VGS = 0, a depletion region forms below the oxide, lowering total capacitance (the curve's minimum).
- Beyond VTH, the inversion (electron) layer forms and capacitance rises back toward Cox — the “bathtub” C-V curve.





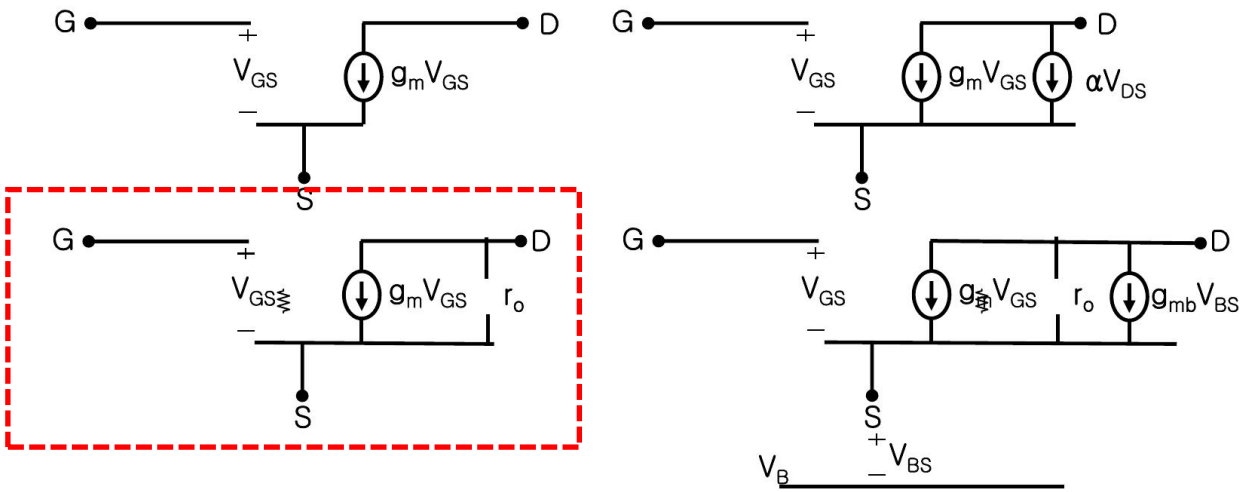
VI

Small-Signal Model

- Four small-signal model variants
- Bulk transconductance (gmb) derivation
- Gate resistance at high frequency

Small-Signal Models of the MOSFET

- Model (1): gmVGS only — simplest, assumes infinite output resistance.
- Model (3): gmVGS in parallel with ro — the standard model for hand analysis, with ro = 1/(λID).
- Model (4): adds gmb·VBS — required whenever the source is not AC-connected to the body (e.g., cascode).



$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\partial I_D / \partial V_{DS}} = \frac{1}{\frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{TH})^2 \lambda} = \frac{1}{\lambda I_D}$$

Bulk Transconductance (gmb) Derivation

- gmb = ∂ID/∂VBS — the body terminal affects ID indirectly, by modulating VTH.
- Chain rule + body-effect equation gives gmb = gm·[γ/(2√(2ΦF+VSB))] = η·gm.
- η is typically 0.1–0.3 — the body behaves as a "second gate" with 10–30% of gm's control.

$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_{TH})\left(\frac{-\partial V_{TH}}{\partial V_{BS}}\right)$$

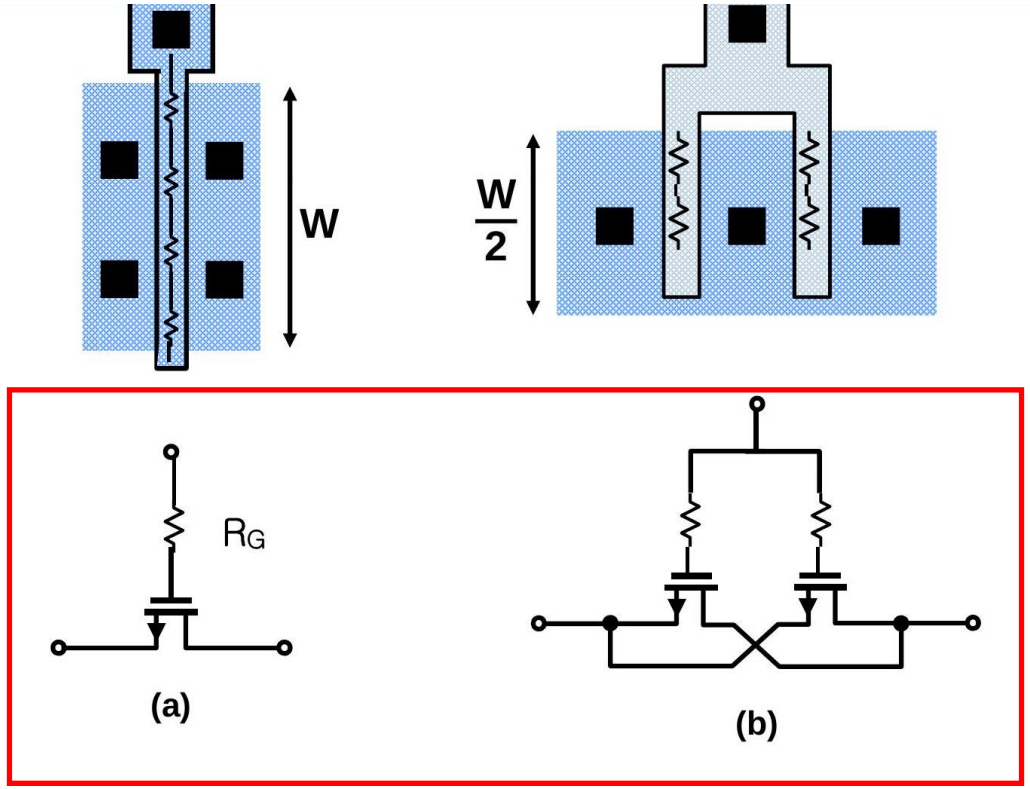
Also,

$$\frac{\partial V_{TH}}{\partial V_{BS}} = \frac{-\partial V_{TH}}{\partial V_{SB}} = -\frac{\gamma}{2}(2\Phi_F + V_{SB})^{-1/2}$$

$$g_{mb} = g_m \frac{\gamma}{2\sqrt{2\Phi_F + V_{SB}}} = \eta g_m$$

Gate Resistance: A High-Frequency Parasitic Effect

- Polysilicon gate has nonzero resistivity, creating a distributed resistance RG along the gate width.
- Contacting the gate from both sides (b) instead of one side (a) reduces the effective resistance to RG/4.
- RG + CGS form an RC network limiting fmax and adding input-referred thermal noise — key for RF/low-noise design.



$\frac{R_G}{2}$